UNIVERSITI TEKNOLOGI PETRONAS
EFB2073 – MICROPROCESSOR & COMPUTER ARCHITECTURE
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MAY 2026



PROJECT FINAL REPORT

SMART MANUFACTURING PRODUCTION COUNTER

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1. Introduction

1.1 Components

Inputs:

- 4 Push Buttons (Start Production | Stop Production | Reject Product | Reset Production)

Output:

- LCD Screen
- 7-Segment Displays
- 3 LEDs (Running | Target Reached | Fault)

Other Components

- Jumper Wires
- Resistors
- Potentiometer

1.2 Project Description

This project involves the development of a production monitoring system for a manufacturing line. Before production begins, a production target (e.g., 20 products) is predefined in the code. When the system starts, the LCD displays “Press the Start Production button to start a new production”. Pressing the Start Production button starts the production process, and each following press simulates the completion of one product. The 7-segment display continuously shows the number of successfully produced products, excluding rejected products.

If a defective product is detected, pressing the Reject Product button increments the reject counter without increasing the production count. The LCD continuously displays the number of rejected products, the number of products remaining to reach the production target (e.g., x/20), and the production efficiency as a percentage, calculated by dividing the current number of accepted products by the total number of products produced.

The Running LED remains ON while production is active. Once the production target has been achieved, the Target Reached LED lights up and the LCD displays “Production Complete”. If the

reject rate exceeds a predefined threshold (e.g., 10%), the Fault LED turns ON and the LCD displays “Quality Warning”, indicating that the manufacturing process requires attention.

The Stop Production button pauses the production process while preserving all production data. The Reset Production button clears all counters and prepares the system for a new production batch.

2. Design Process

2.1 Flowchart Diagram

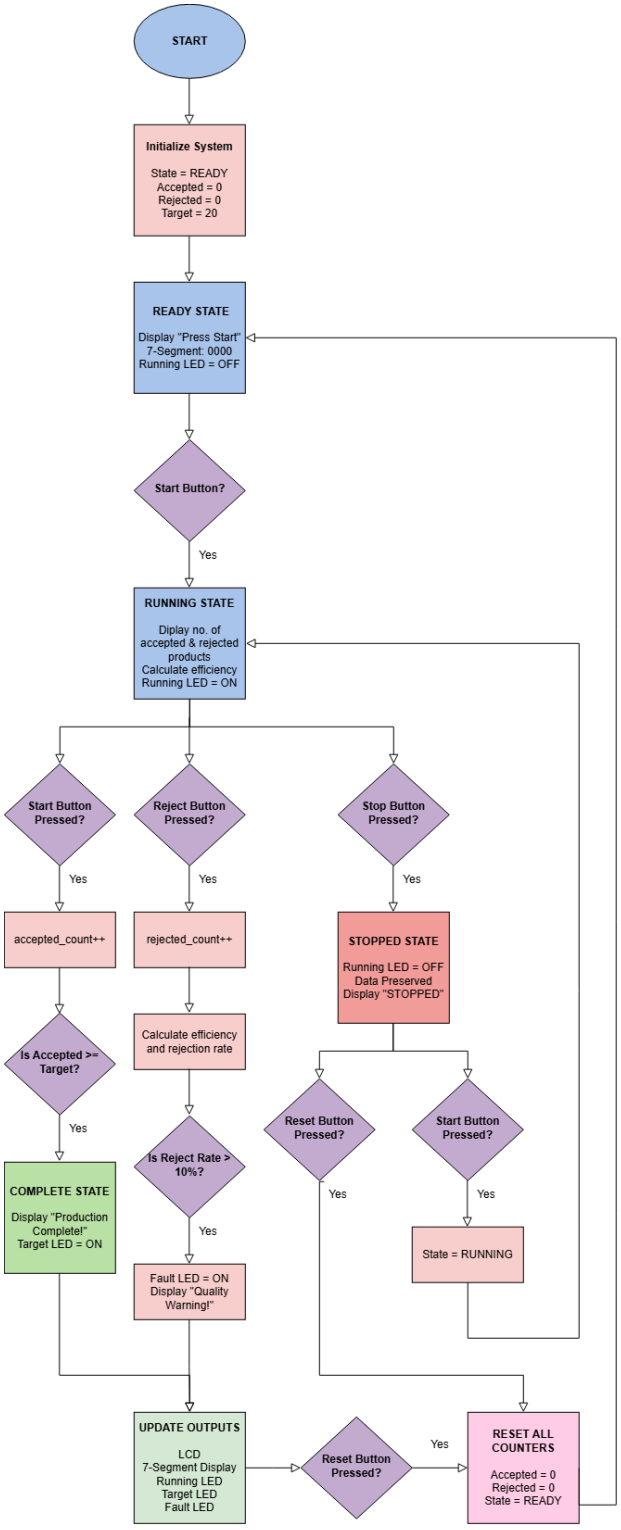


Figure 1: Flowchart

2.1 Diagram Explanation

The design of the Smart Manufacturing Production Counter was developed using a flowchart to represent the system logic and order of operations. The flowchart was used to identify the different operating states of the system, user inputs, processing functions, and corresponding outputs before implementing the system in code.

As shown in Figure 1, the system begins by initializing the production variables: the accepted product count, rejected product count, and predefined production target (set to 20). The system then enters the **Ready State**, where the LCD prompts the user to start production, the 7-segment display is set to zero, and the Running LED remains OFF. When the Start Production button is pressed, the system enters the **Running State** and begins monitoring the production process.

During the Running State, the system continuously waits for an input from the Start, Reject, and Stop buttons. Pressing the Start Production button increments the accepted product count, while pressing the Reject Product button increments the rejected product count without increasing the accepted product count. The system then calculates the production efficiency and rejection rate based on the current production data. If the accepted product count reaches the target, the system enters the **Complete State**, where the LCD displays “Production Complete” and the Target Reached LED is activated.

There is a quality monitoring function that checks whether the rejection rate exceeds the predefined 10% threshold. If the threshold is exceeded, the Fault LED is activated and the LCD displays “Quality Warning” to indicate that attention is needed. The Stop Production button allows the system to enter the **Stopped State**, where the Running LED is turned OFF while all production data is preserved. Production can then be resumed by pressing the Start Production button.

Finally, the Reset Production button clears the accepted and rejected product counters and returns the system to the **Ready State**. This allows the system to start a new production batch while making sure that the previous production data does not affect the new batch.

3. Testing & Troubleshooting

The testing process was initially performed on each component separately before integrating them into the complete system. This allowed hardware and wiring problems to be identified and resolved before they affected the operation of the overall system.

The push buttons, LEDs, LCD, and 7-segment displays were tested individually to verify if they worked. During initial testing, some components were found to be faulty or did not operate as expected. These components were tested and checked to determine whether the problem was caused by the component itself or by the wiring and connections. We had some faulty components and had to quickly find replacements from other Arduino kits or borrow from friends.

One of the main hardware issues encountered involved the 7-segment display. Since the system uses the display to show the number of accepted products, it was necessary to verify that all four digits were functioning correctly. Each digit was tested individually to ensure that it could display the required numbers correctly.

Another issue we faced was during the software development stage, when the ARM Keil Studio development service was discontinued while the project was being developed. This delayed the development and required the project to be completed using the available tools. The code was tested incrementally to ensure that the software logic remained functional as the different system features were implemented.

After the individual components were verified, testing was performed on the complete system. The different operating conditions were tested, including starting production, increasing the accepted product count, registering rejected products, stopping and resuming production, reaching the production target, triggering the quality warning, and resetting the system. The LCD, 7-segment display, and LEDs were monitored during each test to ensure that the outputs corresponded correctly to the system state.

Through this testing and troubleshooting process, the identified hardware and software issues were resolved, and the final system was able to perform the required functions correctly.

4. Circuit Schematic Diagram

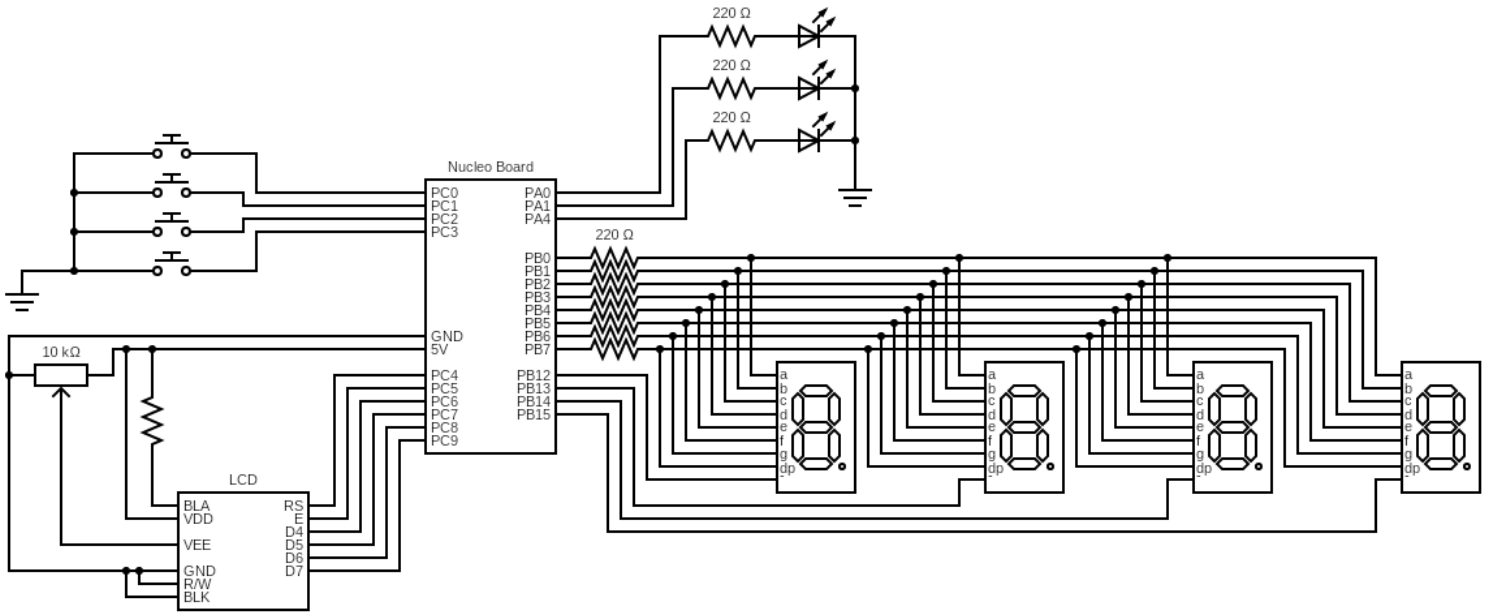


Figure 2: Circuit Schematic